

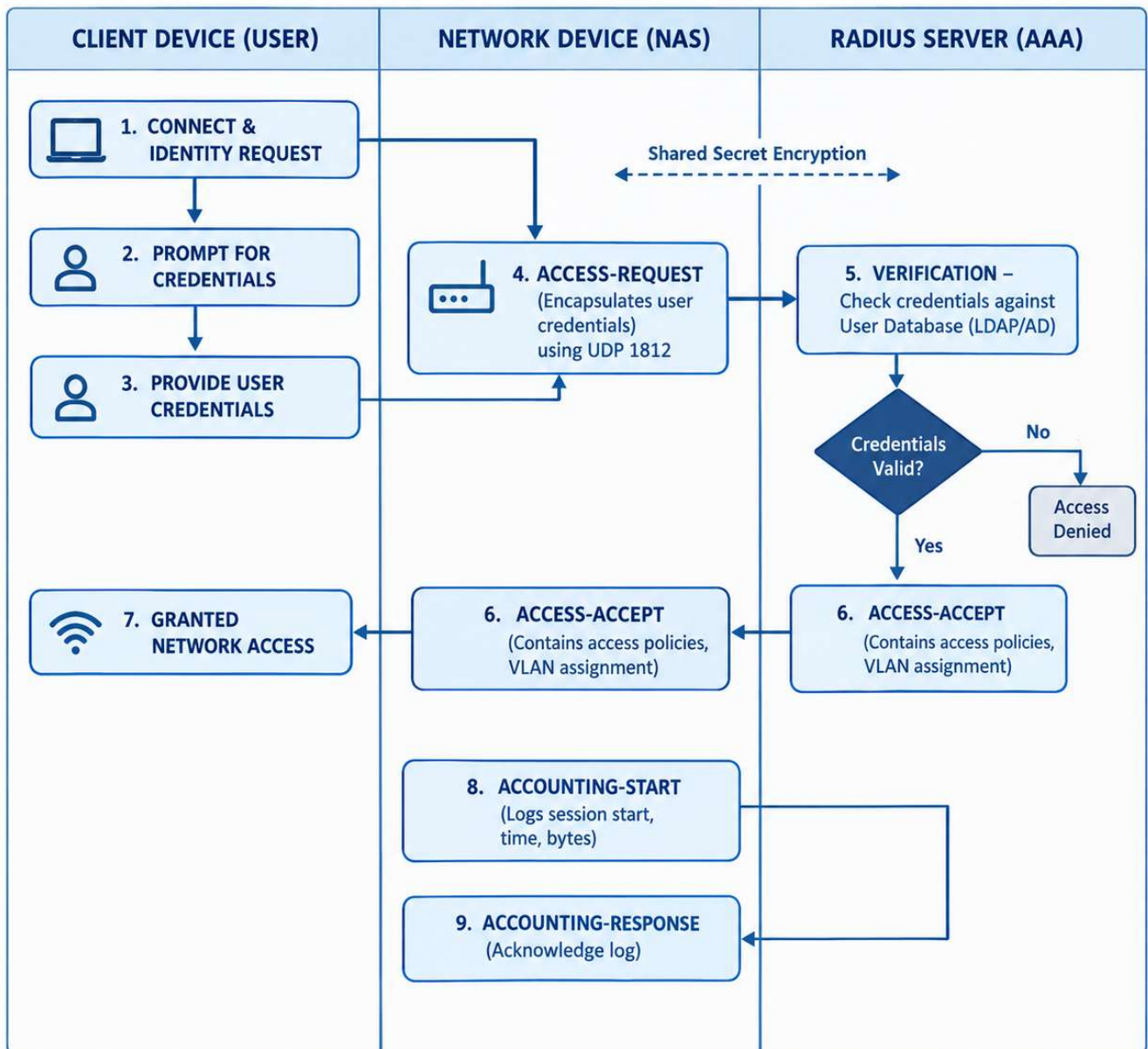
RADIUS & TACACS+

Task 1: Create a comparison table listing at least 4 key features of RADIUS and TACACS+ side by side, highlighting which protocol supports each feature.

Feature	RADIUS	TACACS+	Supported Protocol Details
Transport Protocol & Port	UDP (Ports 1812 / 1813)	TCP (Port 49)	RADIUS: Uses connectionless UDP for speed. TACACS+: Uses connection-oriented TCP for reliable delivery.
AAA Separation	Combines Authentication & Authorization	Separates AAA Functions	RADIUS: Merges authentication and authorization in single response packets. TACACS+: Fully decouples Authentication, Authorization, and Accounting into distinct processes.
Packet Encryption	Encrypts Password Only	Encrypts Entire Payload	RADIUS: Only encrypts the password field in Access-Request packets; headers and body remain plain text. TACACS+: Encrypts the complete packet body.
Granular Command Authorization	Limited / No Per-Command Control	Native Per-Command Authorization	RADIUS: Authorizes session level access

			via attributes (e.g., privilege levels). TACACS+: Validates and logs every individual CLI command typed by an administrator.
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Task 2: Draw a step-by-step flow diagram showing the RADIUS authentication process for a user logging in to a public WiFi hotspot, labeling each communication step between the client, network device, and RADIUS server.



Task 3: List 3 real-world use cases where TACACS+ is preferred over RADIUS, and briefly explain why TACACS+ is better suited for each scenario.

1. **Enterprise Network Device Management**

Why TACACS+:

TACACS+ allows administrators to control **which commands** each user can run. For example, an engineer may be allowed to view settings but not restart a router or change its configuration.

2. **Financial and Payment Networks (PCI-DSS)**

Why TACACS+:

TACACS+ provides **detailed user and command logs** and encrypts more of the communication. This makes it useful where strong security and auditing are required.

3. **IT Support / NOC with Different Access Levels**

Why TACACS+:

TACACS+ can give **different permissions to Tier-1, Tier-2, and Tier-3 staff**. Each person can be allowed to perform only the commands needed for their role.

In simple words:

TACACS+ is preferred when an organization needs detailed control over administrator commands, strong auditing, and different levels of access.

Task 4: Given a scenario where a user tries to log into a network device but fails authentication, describe how RADIUS and TACACS+ would handle the logging and response differently. Hint: Focus on how each protocol manages accounting and command authorization.

	RADIUS	TACACS+
Authentication	Sends an Access-Reject response to the network device.	Sends an authentication failure response.

	RADIUS	TACACS+
Command Authorization	No command authorization takes place because the user did not log in.	No command authorization takes place because authentication failed.
Accounting / Logging	Can record the failed authentication attempt, depending on the RADIUS server/device configuration.	Provides detailed accounting and can record the failed attempt with information such as username, time, source, and status.
Main difference	Mainly focuses on authentication and access control; accounting is a separate function.	Separates authentication, authorization, and accounting, giving more detailed control and auditing .

Task 5: Use ChatGPT or another AI tool to generate a sample RADIUS configuration snippet for a Cisco router, then review the output and list any changes you would make to adapt it for a college campus WiFi setup.

Base Cisco RADIUS Configuration Snippet

! Enable AAA

```
aaa new-model
```

! Configure the RADIUS server

```
radius server CAMPUS-RADIUS
```

```
address ipv4 192.168.10.10 auth-port 1812 acct-port 1813
```

```
key CampusSecret123
```

! Use RADIUS for login authentication

```
aaa authentication login default group radius local
```

! Configure the router's management lines

```
line vty 0 4
```

```
login authentication default
```

```
transport input ssh
```

Changes for a College Campus Wi-Fi Setup

I would make the following changes before using this configuration in a real campus:

1. Use a dedicated RADIUS server

Replace 192.168.10.10 with the actual campus RADIUS server address.

2. Use a strong shared secret

Replace CampusSecret123 with a long, unique secret.

3. Configure multiple RADIUS servers

Use a primary and backup RADIUS server so Wi-Fi authentication continues if one server fails.

4. Use student/staff accounts from a central database

Connect RADIUS to **Active Directory/LDAP** so students and staff can use their normal campus credentials.

5. Use WPA2/WPA3-Enterprise

Campus Wi-Fi should use **802.1X with RADIUS**, rather than a single shared Wi-Fi password.

6. Configure VLAN assignment

RADIUS can assign different VLANs based on the user's identity. For example:

- Students → Student VLAN
- Faculty → Faculty VLAN
- Guests → Guest VLAN

7. Enable accounting

Configure RADIUS accounting to record **login/session start, session end, and usage information**.

8. Restrict RADIUS communication

Allow only authorized network devices such as wireless controllers/access points to communicate with the RADIUS server.

Simple campus Wi-Fi flow

Student → Wi-Fi Access Point → RADIUS Server → Student Database

If the credentials are correct:

RADIUS → Access-Accept → Student gets network access/VLAN

If the credentials are wrong:

RADIUS → Access-Reject → Student is denied access